Patrick Iben, MS

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LEAD BIOSTATISTICIAN

Applied mathematician and biostatistician with seven years in clinical biostatistics and twelve years in research, including nearly four years as lead biostatistician on 12 to 15 early-phase clinical pharmacology studies. Works to eliminate needless ambiguity, risk and suffering in decisions that rest on data. Builds statistical frameworks, AI-evaluation tools and public-data systems with traceable inputs and testable assumptions; independent work includes six research manuscripts, scientific Python software, wearable sensing and 36 public-data consoles. Open to full-time and consulting roles.

AREAS OF EXPERTISE

Clinical Biostatistics | Clinical Trial Design | Statistical Analysis Plans (SAP) | Tables, Listings and Figures (TLF) | Early-Phase Clinical Pharmacology | PK/PD and NCA Analysis | Bioequivalence Studies | CDISC SDTM/ADaM | Sample Size and Randomization | Safety Data Review | Causal Inference | SAS Programming | R and Shiny | Python | Data Quality Control

TECHNICAL SKILLS

Methods: estimands, interim analysis, survival and longitudinal analysis, missing-data imputation, cluster randomized trials, causal inference under interference (spillover and contamination), consilience testing across data sources, simulation, permutation inference, calibration and decision curves.

Tools: R (pharmaverse and Shiny), Quarto, SAS, Python, SQL, JavaScript, Git, Docker, LaTeX, Veeva Vault and high-performance computing. Data provenance, automated quality control (QC), browser testing, public API integration and numerical validation.

Therapeutic areas: cardiovascular, dermatology, immunology, neurology, oncology, psychiatry and respiratory.

PROFESSIONAL EXPERIENCE

Biostatistician II | **Syneos Health** | Remote | December 2021 to October 2026

- Progressed from support statistician to lead biostatistician within a year of joining; from December 2022, led 12 to 15 early-phase clinical pharmacology studies across therapeutic areas, including first-in-human, single and multiple ascending dose (SAD/MAD), crossover, bioequivalence and bioavailability designs, guiding statistical decisions through reporting.
- Owned protocol statistical sections, randomization schedules, statistical analysis plans and mock tables, listings and figures (TLFs); created or reviewed CDISC programming specifications for analysis datasets and outputs.
- Performed pharmacokinetic/pharmacodynamic (PK/PD) and non-compartmental (NCA) analyses, wrote SAS review programs for quality control and reviewed all TLFs before delivery; contributed to clinical study reports (CSRs).
- Reviewed interim study data, annotated case report forms (CRFs) and database designs against analysis requirements, ensuring collected data would support the planned analyses.

Associate Biostatistician | **Emmes** | Rockville, MD | September 2019 to December 2021

- Led and supported federally sponsored Phase 1 and Phase 3 opioid-use-disorder trials; wrote protocol and analysis-plan sections and the statistical sections of final study reports and of Data and Safety Monitoring Board (DSMB) reports that informed safety oversight.
- Developed dynamic progress reporting for ongoing protocols, giving study teams timely visibility, and SAS/R tools for data-management and analysis problems.
- Authored the data extraction and normalization programs that created a multi-clinic electronic health record (EHR) study dataset, credited by name in a 2024 *Addiction* paper; gave statistical support to external research partners.

Research Assistant and Teaching Assistant | **University of Colorado Anschutz** | Aurora, CO | August 2016 to July 2019

- Harmonized multi-state hospital databases containing millions of records and hundreds of derived variables as biostatistics research assistant for the Data to Value nursing econometrics initiative at the University of Colorado College of Nursing, supporting pediatric nursing-cost research and reuse in later studies.
- Advised about three doctoral nursing candidates on study design and sample size; taught graduate biostatistics review sessions and office hours for three years in courses including Biostatistical Methods and Advanced Data Analysis.

Earlier research roles | 2014 to 2016

Postsecondary and Policy Analysis Fellow, State Higher Education Executive Officers; Data Analysis Fellow, Yale School of Child Development; Senior Data Analyst, Southern Connecticut State University Office of Assessment and Planning.

SELECTED PROJECTS

Trial biostatistics toolkit and eCRF pipeline apps

A public toolkit for biostatisticians: SAS and R templates, dashboards and quality-control checks from protocol statistical sections to report text. It runs locally, so client data stay on the analyst's machine; the example studies are synthetic.

- Created a complete worked example of a bioequivalence study: one synthetic trial taken from sample size and randomization through the analysis plan, analysis datasets, pharmacokinetic analysis, tables and report text, all regenerated by a single script.
- Designed safety-monitoring tools for first-in-human studies: liver and cardiac screening rules, a Safety Review Committee packet and an interactive dashboard that runs in the browser to streamline safety review.
- Engineered a quality-control harness that recomputes key results a second way and is tested against planted errors.
- Wrote SAS and R program libraries for standard tables, listings and figures across common early-phase designs, plus simulation studies of two regulatory checks.
- Released four browser-based apps for reviewing clinical trial data (medical coding, de-identification, participant risk and central monitoring) and published their measured limitations.

Public Data Consoles and Coastal Water Resilience

- Built 36 browser-only decision-support consoles on public data feeds and dated snapshots across hazards, public health, access to care and markets; each labels every input as measured, modeled or demo so users can judge evidence quality.
- Unified two original research methods in Coastal Water Resilience: a ten-metric diagnostic suite from contamination-resilient cluster randomized trial design plus directed source/sink topology accounting, applied to live NOAA and National Weather Service forecasts across 131 published wastewater lift stations to drive projected impacts and response plans.

Autonomic Telemetry and Resilience Telemetry

- Prototyped a personal physiological-arousal biofeedback system combining heart rate, heart-rate variability, EEG and covariance distance against a personal reference, on live Muse S and Hume Band streams via a local Mac-to-iPhone bridge.
- Applied original directed-contagion research to cross-asset market volatility: spillover networks, transmitter stability and historical replay, tested against held-out baselines that track the loudest and most persistent assets.

Consilience Engine

- Packaged gene-set analysis as a Python library and command-line tool using InterPro, Human Protein Atlas, STRING and Gene Ontology; tested with permutation nulls on a 410-gene nervous-system map and a 132-gene immune replication.

EDUCATION AND TRAINING

MS, Biostatistics, Data Science Analytics minor | University of Colorado Anschutz

BS, Mathematics, Applications of Mathematics | Southern Connecticut State University | Magna cum laude

High-Performance Computing Training and Certification | University of Colorado Anschutz

PUBLICATIONS AND MANUSCRIPTS

Submitted manuscripts, with review status as of 1 November 2026:

- **Contamination-Resilient Cluster Randomized Trial Design under Spatial Interference: A Simulation Study of Contamination Control.** *Statistics in Medicine*, submitted 19 October 2026. Preprint doi:10.5281/zenodo.23486176.
- **Auditing preprocessing-sensitive allele rankings in predicted chromatin contact maps.** *BMC Bioinformatics*, submitted 29 October 2026. Integrates an earlier preprint: What do within-site comparisons of AlphaGenome Atlas predictions measure? An audit at ClinVar sites in 11 genes. doi:10.5281/zenodo.23543154.
- **A false floor: Refuting our own information-limit claim in a lethal identification task.** *PLOS ONE*, submitted 22 October 2026. Preprint doi:10.5281/zenodo.23519299.
- **Anticipating Network Contagions: When Directedness Outperforms Connectedness.** *Applied Network Science*, with editor.
- **Which data modality recognizes a gene set? A domain-agnostic consilience engine.** *PeerJ*, in peer review.
- **No emotion-specific image-computable code in the human left amygdala.** Peer Community In, preprint review. Preprint doi:10.5281/zenodo.21422754.

Published work:

- **Acknowledged contributor:** Osterhage KP, Hser YI, Mooney LJ, et al. Identifying patients with opioid use disorder using International Classification of Diseases (ICD) codes: Challenges and opportunities. *Addiction*. 2024;119(1):160-168.
- **Co-author:** Ben-Avie M, Rosenwald L, Depastino R, Iben P. How can we help young adults with autism thrive in the workplace? *The Conversation*, 2016.